# Sales Data Analysis & Business Insights System

Week 3 Python & Data Analysis Project

## 1. Project Objective

The objective of this project is to examine historical sales data, identify data-quality issues, analyze product and regional performance, investigate sales trends, and provide evidence-based recommendations.

## 2. Dataset Overview

The current DataFrame contains 5,179 rows and 12 columns. The dataset includes order identifiers, dates, products, quantities, prices, regions, customer identifiers, payment methods, order statuses, and derived sales values where available.

## 3. Data Cleaning and Quality

| Data-quality check | Current result |
|---|---|
| Missing product names | 130 |
| Missing regions | 1 |
| Missing quantities | 0 |
| Missing prices | 177 |
| Missing customer IDs | 166 |
| Invalid or missing dates | 321 |
| Duplicate rows | 3 |

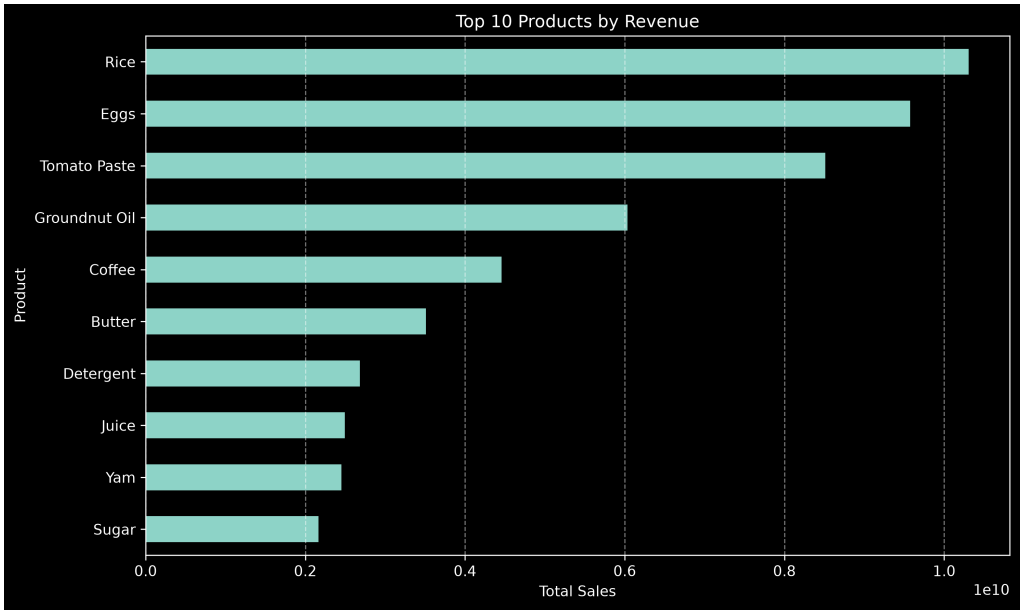
Data preparation included investigating missing values, standardizing categorical values, checking numerical fields, converting dates, and reviewing duplicate records. Missing product names and regions should not be replaced with invented information.

## 4. Product Performance

The highest-revenue product in the current calculation is Rice, with recorded sales of 10,305,941,400.00. Revenue rankings can help identify products for further inventory and demand analysis, but revenue alone does not measure profit.

| Product | Recorded sales revenue |
|---|---|
| Rice | 10,305,941,400.00 |
| Eggs | 9,574,530,200.00 |
| Tomato Paste | 8,512,125,900.00 |
| Groundnut Oil | 6,033,568,600.00 |
| Coffee | 4,455,750,700.00 |

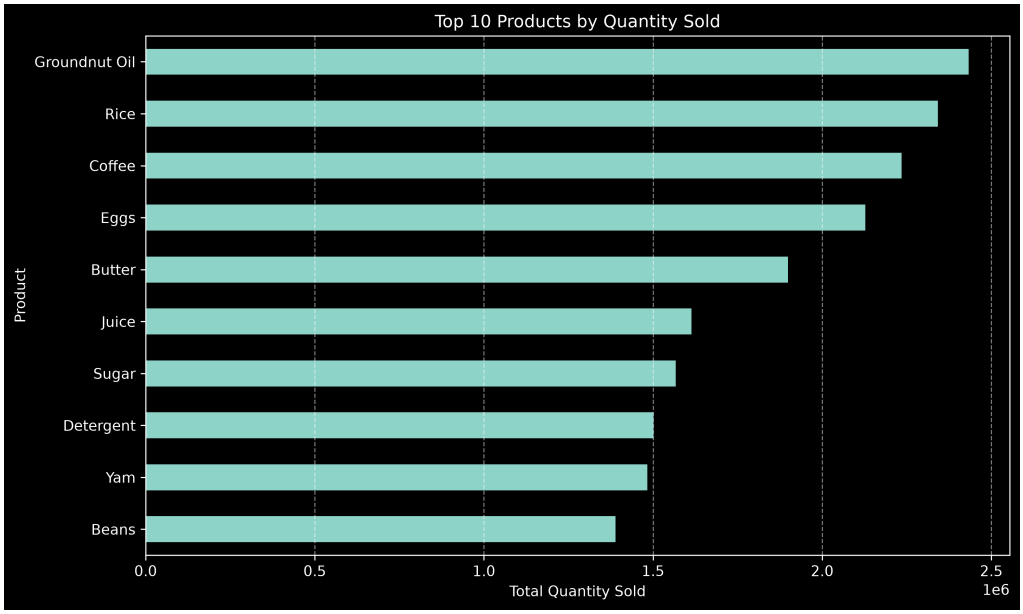
## Chart 1: Top 10 Products by Revenue




This chart compares the ten products with the highest recorded revenue.

## Chart 4: Top 10 Products by Quantity Sold

| Product | Quantity sold |
|---|---|
| Groundnut Oil | 2,432,700.00 |
| Rice | 2,342,280.00 |
| Coffee | 2,234,450.00 |
| Eggs | 2,127,000.00 |
| Butter | 1,899,170.00 |



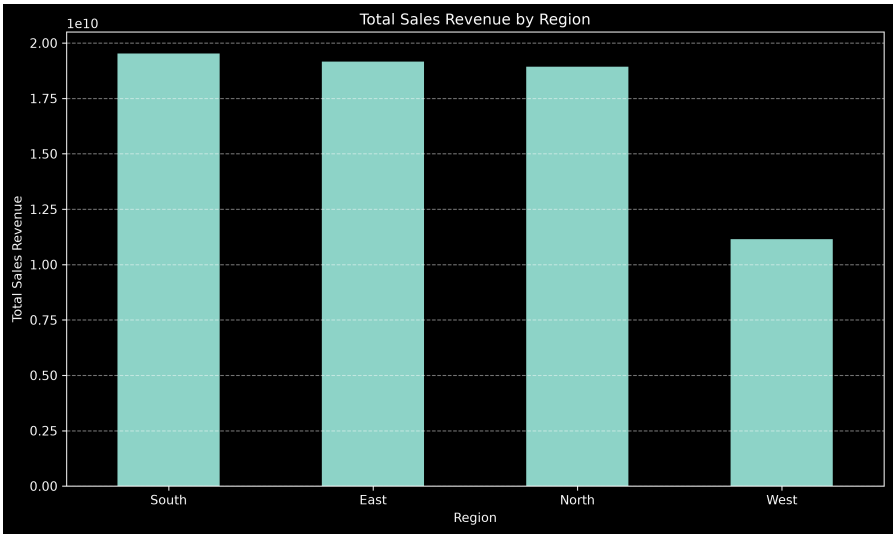

This chart ranks products by their total recorded quantity sold. A product with high quantity sold may not generate the highest revenue because unit prices differ.

Recommendation: review stock availability, quantities sold, unit prices, and product costs before deciding which products to prioritize.

## 5. Regional Sales Performance

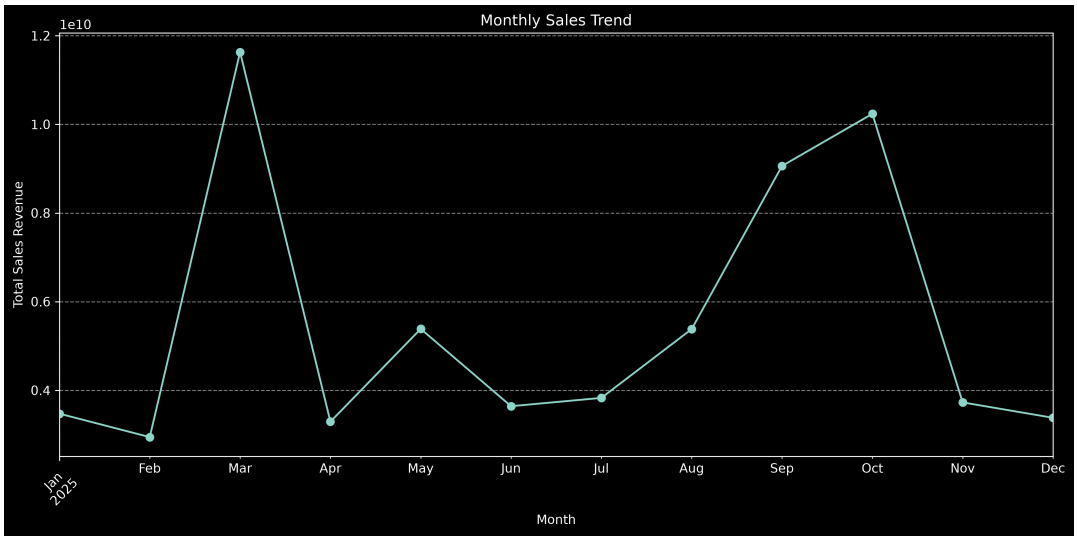
The highest-revenue region in the current calculation is South, with recorded revenue of 19,526,382,400.00.

| Region | Revenue | Unique order IDs |
|---|---|---|
| South | 19,526,382,400.00 | 1,315 |
| East | 19,168,089,300.00 | 1,250 |
| North | 18,929,339,250.00 | 1,280 |
| West | 11,146,517,800.00 | 1,305 |




## 6. Monthly Sales Trend

The highest-revenue month in the current calculation is 2025-03, with recorded sales of 11,629,824,400.00. This identifies a strong month but does not by itself establish a seasonal pattern.

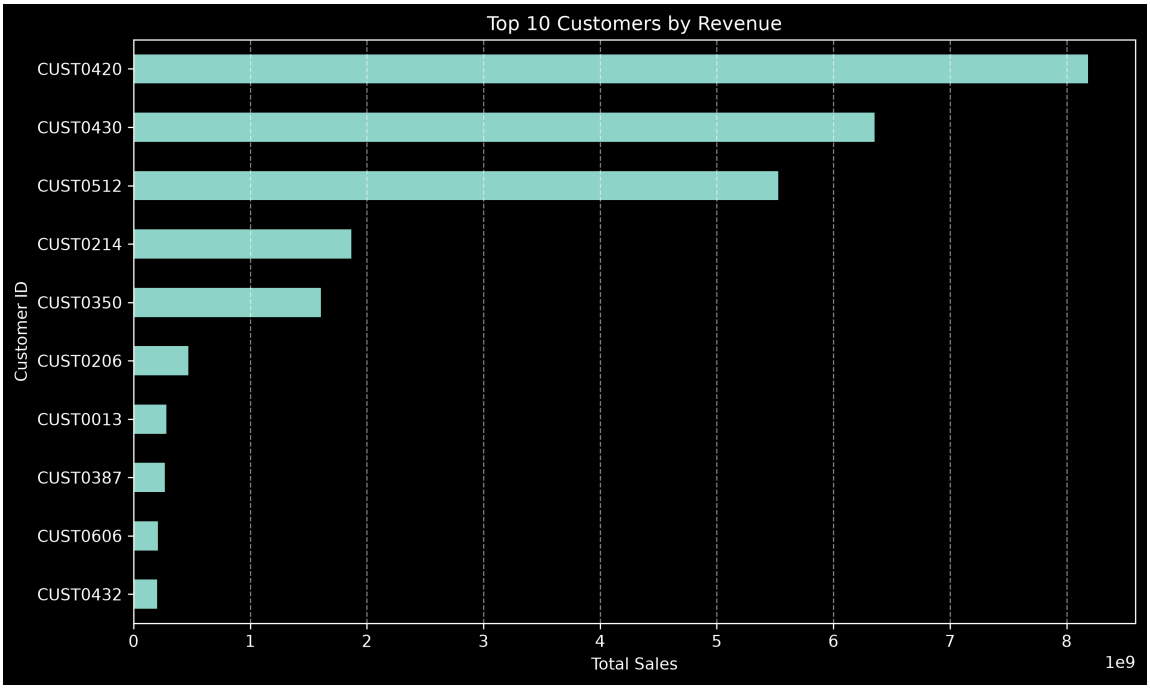



Recommendation: monitor monthly revenue and investigate significant increases or decreases using order counts, quantities, prices, and unusual transactions.

## 7. Customer Revenue and Order Analysis

The highest-revenue customer in the current calculation is CUST0420, with recorded revenue of 8,181,534,500.00. This result should be checked for unusually large transactions before drawing conclusions.

| Customer ID | Recorded sales revenue |
|---|---|
| CUST0420 | 8,181,534,500.00 |
| CUST0430 | 6,353,695,700.00 |
| CUST0512 | 5,526,426,600.00 |
| CUST0214 | 1,867,565,800.00 |
| CUST0350 | 1,605,135,100.00 |




This chart identifies the ten customers with the highest recorded sales revenue. These results can guide further customer-retention analysis and investigation of purchasing patterns.

## 8. Analytical Method

The analysis used Pandas grouping, aggregation, sorting, date conversion, and unique order counts to compare performance across products, regions, months, and customers.

```
region_sales = df.groupby('region')['total_sales'].sum()
product_sales = df.groupby('product')['total_sales'].sum()
product_quantity = df.groupby('product')['qty'].sum()
customer_sales = df.groupby('customer_id')['total_sales'].sum()
monthly_sales = df.groupby(
    df['order_date'].dt.to_period('M')
)['total_sales'].sum()
```

These summaries provide a repeatable way to examine sales patterns. The detailed calculations and cleaning decisions are documented in the Jupyter Notebook.

## 9. Key Business Insights

**1. Regional performance:** Compare revenue and order volume to understand differences in regional performance.

**2. Product revenue:** Identify high-revenue products and investigate their demand, availability, and costs.

**3. Product quantity:** Compare quantities sold with revenue to understand which products sell most frequently.

**4. Monthly trends:** Investigate high- and low-revenue months, checking order counts and unusual transactions.

**5. Customer contribution:** Validate major customer totals and compare purchasing frequency and average order value.

**6. Order volume:** Compare unique order counts with revenue to distinguish transaction volume from transaction value.

## 10. Recommendations

Validate missing and unusual records before making business decisions. Monitor regional and product performance, review customer purchasing patterns, and investigate monthly changes in sales. Compare product quantity sold with revenue, and obtain product-cost information before evaluating profitability.

## 11. Conclusion

This project demonstrates how Python and Pandas can be used to inspect, prepare, summarize, and analyze sales data. The results can help guide further business investigation, provided the final figures are validated against the cleaned dataset. Revenue should not be interpreted as profit, and observed differences should not be treated as explanations of cause.